

Significance of the Focused Program of Research

If students are to be successful in school, at work, and in their personal lives, they must learn to write (Graham, 2019). Employer surveys consistently highlight effective written communication as a top skill sought in job candidates (National Association of Colleges and Employers, 2018; National Commission on Writing, 2004). Yet students do not enter college as strong writers (ACT, 2016), nor do they leave it writing well enough for their employment (Finley, 2021), so employers spend over \$3 billion each year on remedial writing training (Moore, 2016).

Postsecondary educators are faced with resource constraints that make it difficult to provide the needed quality instruction for writing (Crespín-Trujillo & Hora, 2021; Ran & Sanders, 2020). Writing is a process-based skill that requires iterative, detailed, and individualized feedback and guidance (Graham, 2019). Yet, writing faculty often need to teach multiple classes with limited resources, which makes it challenging, if not entirely infeasible, to provide timely, high-quality, and personalized feedback to each student along their learning process.

These challenges remain true even in the face of ubiquitous generative artificial intelligence (AI). The best writers without generative AI are also the best writers *with* generative AI, because they both understand what specifically to ask the AI for (e.g., knowledge of genre, audience) and they recognize good writing and can edit as needed. Strong writing skills enable students to critically engage with AI-generated text, refine AI output, and use AI as a tool rather than a crutch. Writing will remain a fundamental skill for other reasons as well. Writers use this versatile skill to think through complex information and learn content across the curriculum (Graham, 2018; Graham & Hebert, 2011). While generative AI has a role to play in many of these uses, it does not supplant the need for postsecondary students to have sufficient writing skills to take advantage of AI's affordances and not fall prey to its limitations.

Generative AI tools provide a powerful means of scaffolding the development and practice of writing: They can help ask the right questions, point students toward counterarguments that should be addressed, offer suggestions for getting started, support task management, provide model texts, and give coherent feedback on student writing 24/7 (AlAfnan et al., 2023; Cummings et al., 2024; Meyer et al., 2024; Tate et al., 2024). In theory, generative AI can complement faculty efforts, rather than replace them. Faculty can focus on higher-order writing development while students benefit from timely, detailed, and personalized feedback that guides them through the writing process. Faculty may also benefit from timely information on student progress and needs that AI might make available.

Yet since these tools can quickly generate plausible texts on almost any topic, students may be tempted to use these tools to write texts that they themselves have been assigned, shortchanging students' own development as writers (Lee et al., 2025). Weak writers may be most tempted to use AI in unproductive ways, especially if they have not developed the AI literacy to use them responsibly.

All of these challenges are more acute at our nation's community colleges, which enroll around 40% of all undergraduates (National Center for Education Statistics, n.d.). As a result of their open-access missions, low tuition, and flexible course scheduling, community colleges serve a disproportionate number of first-generation college students, low-income learners, working adults, and veterans. This broad access makes community colleges a key driver of social mobility (Bailey et al., 2015). However, students in community college have less academic preparation than their counterparts in four-year schools and are much more likely to fail or drop out. And failure in first-year composition courses reduces a student's chance of graduation by 38%, more than any other first-year class (Garrett et al., 2017). Communication college

instructors more frequently serve in part-time positions at multiple institutions which creates significant time constraints in providing personalized feedback to students (Crespín-Trujillo & Hora, 2021; Ran & Sanders, 2020; Xu & Ran, 2021). In addition, instructors in community colleges have less administrative and technology support and less access to professional development opportunities than their peers in four-year institutions. And while traditional writing support models, such as writing centers or peer tutoring have been used by colleges to provide writing assistance, these resources are often limited and understaffed at community colleges.

Generative AI shows promise of providing some of the necessary personalized instruction and scaffolded support in a manner that supplements, not supplants, the classroom instructor, and may be especially valuable in community colleges where additional assistance is most needed but hard to access (e.g., working students may not have time to access the writing center during its business hours). However, there are many challenges to the use of generative AI including difficulties of instructors and institutions in understanding, evaluating, and selecting among the large number of generative AI-based writing tools; a dearth of information on evidence-based ways for implementing these tools and data on their impact; and guidance as to how to navigate the complex privacy, bias, and ethical risks of generative AI. The WRITE AI Center will tackle these challenges through addressing the following three research questions:

1. What are the most widely used generative AI tools for teaching and learning of postsecondary writing and what are their relevant features for instruction?
2. How can a generative AI writing tool best be further developed and deployed to support writing instruction in community college and what is its impact on student and teacher learning outcomes related to writing quality, AI literacy, self-efficacy, and persistence?
3. What are the best approaches for mitigating the risks, biases, and ethical challenges of incorporating generative AI in postsecondary writing instruction?

The WRITE AI Center will focus on the problem of practice of providing sufficient personalized writing instruction and scaffolded support to ensure postsecondary students gain the necessary writing skills and AI literacy to succeed in the workforce.

Theoretical and Empirical Background

There is abundant research suggesting that personalized instruction; scaffolded support; strategy instruction; and prompt, actionable feedback in a supportive environment from a trained instructor or tutor are invaluable for helping students learn to write (Graham, 2019; MacArthur et al., 2015). Personalized feedback on students' writing plans and drafts is recognized as one of the most effective ways of helping improve students' writing ability (Graham et al., 2011), especially when that feedback is delivered during interactive one-to-one writing conferences (Bayraktar, 2012). Yet instructors are often unable to give the detailed feedback and guidance they know students need, especially on early drafts of student writing, and are even less likely to be able to individually assist students in the prewriting stages of writing.

Because of the time limitations on instructors in providing sufficient guidance and feedback on writing, there has long been interest in developing software for this purpose, yet decades of research and development on automated writing evaluation systems have yielded limited results (see discussion in Ifenthaler, 2022). While automated systems can give reliable numerical scores on writing by drawing on machine learning techniques to compare essays to those on the same topic previously scored by humans (Huawei & Aryadoust, 2023), prior to the advent of generative AI, they have been unable to provide helpful formative feedback on student

writing (see discussion in Wilson & Macarthur, 2024) or provide tailored interactive coaching in earlier stages of writing.

In contrast, a strength of large language models is that they can flexibly provide feedback and coaching in a wide range of complex learning domains. There is growing evidence that generative AI is very good at providing feedback on student writing as well as on guiding students throughout the writing process (Benharrah et al., 2024; Hutt et al., 2024; Stamper et al., 2024). In one of our early studies of the topic, using the GPT-3.5 platform that has been far surpassed in ability by more recent models, we found that the quality of generative AI feedback on student essays approximated that of feedback provided by expert instructors (Steiss et al., 2024). A growing body of work also indicates that interacting with an LLM helps people write better (Noy & Zhang, 2023). Yet these studies have been conducted in isolation and may not account for the potential disadvantages in using generative AI to scaffold the writing process. For example, a skilled instructor knows how to adapt the kinds and amounts of scaffolding provided so that students can continue to advance through their “zone of proximal development” (Vygotsky, 1978), thus reaching increasing levels of achievement. In contrast, without careful human control of what kinds of scaffolding are provided when, generative AI can be overused by students in ways that substitute for students’ own deep thinking, skill building, and learning. Additional risks include biases inherent within tools, threats to student privacy, and ethical issues related to authenticity of student work.

Given this balance of affordances and risks, ensuring that both instructors and students become AI literate is key for supporting productive and responsible use of generative AI for writing instruction. Generative AI tools are rapidly being integrated into classrooms, yet both teachers and students often lack the necessary AI literacy to effectively and ethically engage with these technologies. Our research on classroom uses of technology, and in particular AI, suggests that the value of these tools requires understanding of their strengths and limitations; deploying them in a way that maximizes the former and minimizes the latter (Warschauer, 2003, 2006, 2011). AI literacy, or the set of skills to critically use and evaluate AI (Long & Magerko, 2020; Ng et al., 2021), is becoming increasingly important in order to inform implementation of AI tools and to prepare students not only for classroom use, but also for futures and careers where generative AI will likely have an impact (Wong et al., 2020). One way to develop AI literacy is to design classroom AI applications not as the “solution” to AI in the classroom, as many commercial applications are advertised, but with the learning of AI literacy authentically built-in.

Existing research has begun to identify design patterns and features that can help students learn about AI through their interaction with AI, such as transparency (Williams et al., 2019). Users should be able to see where the app uses AI and how the AI interacts with the user (see discussion of AI transparency and critical AI literacy in Long & Magerko, 2020). Additionally, it is important to scaffold and guide the students’ interaction with AI. Apps like ChatGPT lack this, simply providing an empty sandbox to the student, which leads to missed opportunities for learning (Belghith et al., 2024). Researchers have identified the importance of guiding and structuring the students’ interaction with AI in the classroom (e.g., Bilstrup et al., 2024). While Han et al. (2024) identified educator oversight and control as an important design consideration, they also identified how this can be a tension with providing flexibility, as student agency is also of great importance to foster motivation and to provide them with the opportunity to think critically about their work. Students need to develop agency not only with respect to their own data, but also in critically evaluating and reflecting on their use of AI. Ensuring appropriate AI literacy will be critical for developing an AI-enabled workforce and citizenry.

Finally, though there is great potential and great challenges in using AI for writing and AI literacy instruction, colleges lack evidence-based frameworks for evaluating the impact of tool use on student learning over time, or in selecting among AI tools to deploy for this purpose. The Center will help fill this gap, both through new evaluative research, and through widely accessible training opportunities for researchers to gain skills, knowledge, and resources from national experts in evaluation research on generative AI.

The Center’s leadership team, which has deep experience in the teaching and learning of writing, strategies for integrating educational technology in community colleges, and the development and use of tools and resources for improving undergraduate writing with generative AI, will conduct research, training, and leadership activities to support effective and ethical use of generative AI in postsecondary writing instruction.

The Tool: PapyrusAI

PapyrusAI is a generative AI platform for the teaching and learning of writing and AI literacy appropriate for use in developmental, foundational, and disciplinary writing courses in postsecondary education. It was first developed by the UCI PI team in 2023 with grant funding from the National Science Foundation’s Improving Undergraduate STEM Learning program. It has been further revised and improved with grants from the UCI Beall Applied Innovation, The Learning Agency, and the California Education Learning Lab so that it is broadly suitable and effective for writing and writing-intensive courses across the University of California, California State University, and California Community College systems. PapyrusAI has been successfully piloted in composition, writing across the curriculum, ESL, and English classes. The PI team has complete, persistent access to embedded data of teacher and student use of the tool, and also the ability to flexibly and quickly modify the tool in response to what we learn through the development and pilot study, facilitating iterative improvement.

What is PapyrusAI?

PapyrusAI is a web-based tool built on a commercially available generative AI model (currently, GPT-4o) that:

- Leverages advances in generative AI to provide personalized, just-in-time writing assistance.
- Builds in research-based best practices for the teaching and learning of writing through embedded prompts that guide student-AI interactions in productive ways and reduce the likelihood of students’ off-loading important skill building.
- Amplifies teacher pedagogical and contextual knowledge by allowing instructors to adapt and customize the guided interactions and embedded prompts to meet the needs of their students and support course learning objectives.
- Reduces the need for instructors to become AI experts by providing professional development and instructional resources to support students’ AI literacy development.
- Facilitates students’ use of generative AI tools through human-centered use, providing practice interacting in a scaffolded environment and preparing them for future use of commercial tools.
- Provides enhanced privacy protection compared to individual access options.

The PapyrusAI tool helps students improve their writing and digital literacy by allowing them to interact with generative AI in a “walled garden” that provides access to AI in a structured, scaffolded, and bounded way. PapyrusAI interfaces with existing large language models, such as OpenAI’s GPT-4o, in a way that supports both writing development and AI

literacy, with a particular focus on helping them effectively plan and revise their writing. Our work to date has prioritized these underutilized parts of the writing process of developing writers (Graham & Perin, 2007; McCutchen, 1996; Scardamalia & Bereiter, 1987) both because of students' frequent lack of attention to them, and because even if generative AI is drafting a text the planning process supports effective AI prompting and revision of AI output is necessary to corroborate, interrogate, and refine the AI-generated text.

Instructors select from dozens of research-based, tested prompts to support all phases of the writing process. They may enable any or all of the prompts (see Appendix C-1 for current prompt library) over time as they deem appropriate and have the ability to revise the prompts to align it with the class (e.g., referencing specific vocabulary or strategies stressed in the class in the AI-student interactions, or having a prompt focus on specific areas where development is needed, such as logical transitions, voice, or audience awareness).

Students interact with the AI in “conversations” that mimic the format of tools like ChatGPT or CoPilot (see Appendix C-2). The questions asked by the AI explicitly model the steps the students should take when reviewing their own writing, for example, with the intention of providing practice and increasing the transfer of these strategies and skills to the students' independent writing revision. These embedded writing strategies support students' self-regulation by giving students a systematic way to approach the complex task of writing (MacArthur et al., 2015). Concurrently, students gain a facility with harnessing AI for their own authentic purposes and become prepared for eventual use of unbounded AI through practice within the scaffolded PapyrusAI environment.

After the initial conversational turn, students are able to ask questions for clarification or ask the AI to expand upon something. We are finding this to be an important part of students' gaining facility with using AI effectively and maintaining agency in their writing (and AI use). Students need to be explicitly taught to “talk back” to the AI and interrogate its output, asking for elaboration, clarification, expansion, or revision for example (this is consistent with evolving research on AI conversations, see Shah et al., 2024; Wei et al., 2022). For some students, this is uncomfortable, while others lack the language in which to push back to improve the AI output. We have instructional materials to help with this issue, including sentence starters to provide suggested phrases (Appendix C-3). This type of instructional scaffolding is frequently used to support English learners, and it will also allow us to develop AI literacy through exposure to domain-specific vocabulary in future iterations. Instructors can monitor all AI-student interactions through the report function (see Appendix C-4). The development study discussed below will help us enhance teacher reports to provide actionable information to guide instruction.

Core Features

PapyrusAI (including its essential practices and structural elements, Blase & Fixen, 2013) has three core features. The first core feature is a ***web portal*** (a web-based tool; see Appendix C-5) that allows students to interact with a large language model (LLM) via its Application Programming Interface (API), without requiring a student account or logging into the LLM directly and without providing the LLM access to the interactions for training purposes. The web portal allows the instructor to organize and select the types of AI-student interactions enabled on the tool by enabling the embedded prompts (discussed below). The web portal allows the students to select from the enabled prompts and interact with the LLM in a conversational interface that mimics current commercial tools. Students can also view and download the complete texts of these interactions. The development study will incorporate design changes desired by our participating instructors and students, as well as implement a multi-agent structure

that will allow us to enhance the tools features as described below.

Our design principles center the Design Principles for Generative AI Applications (Weisz et al., 2024; see Appendix C-6): design responsibly, design for mental models, design for appropriate trust & reliance, design for generative variability, design for co-creation, and design for imperfection. PapyrusAI provides learning opportunities so that students actually *experience* the power—and limitations—of AI (e.g., Druga et al., 2021; Ng et al., 2021) in support of their authentic classwork. By interacting with AI through a guided process, students gain valuable knowledge related to AI's strengths, weaknesses, biases, and uses, and skills for prompting, incorporating, and interrogating its output. This invaluable AI literacy will be relevant in their future study and careers. Repeated interactions with the tool provide students with the experience and confidence to interrogate other forms of AI in the future, knowing that the AI will make mistakes and that the student's input and refinement of the questions asked of the AI significantly impacts the quality and relevance of the output. We also design with a Teacher-in-the-Loop focus, an adaptation of human-in-the-loop design principles (Calo & Maclellan, 2024; Chen et al., 2023; Soudi et al., 2023) that values and in fact amplifies the pedagogical and context knowledge of the classroom teacher in the design process.

Embedded prompts, the second core feature of PapyrusAI, provide instructions that begin the student-AI interaction with specific information and override the AI's general mode of interaction by telling it how it should and should not interact with students (e.g., *not* to write for the student). The embedded prompts have been based on the best practices for the teaching and learning of writing and one-on-one writing conferences, where students are queried about their ideas and continually pushed further in their thinking, rather than being simply told what to do. The prompts will also be refined over time with suggestions from our Advisory Board writing experts, MacArthur and Graham; participating instructors about their students' needs and abilities; as well as data about how students actually use the tool and our testing of various prompts during development. The embedded prompts (Appendix C-7) are currently in the categories of planning, drafting, revising, presenting, and miscellany, as well as prompts for instructors' instructional purposes. Each of these categories contain 5-7 specific prompts developed and tested by our research team.

Our prompts generally follow a consistent structure that we have tested and found most successful for quality AI output (see Appendix C-8): We begin with the applicable persona (the role we want the AI to play, such as a teacher or writing coach), goal of the interaction (e.g., provide feedback on genre), steps for the AI to follow, and finally introduce any constraints (e.g., don't revise the student's writing for them). Our instructional resources for both instructors and students provide training on how to successfully develop prompts of their own. We also note that as suggested by Simkute et al. (2024), our embedded prompt text is visible (Appendix C-9) so that students can build accurate mental models about how the AI functions and what effective prompts look like.

The third component of PapyrusAI is a set of ***instructional resources*** (Appendix C-10), including written materials, professional development (PD) delivered primarily synchronously and remotely, and videos, that guide instructors and students on how to use the tool effectively to improve student writing and AI literacy. Instructional resources will ensure that students understand the basics of AI and its functions, strengths, weaknesses, and biases (Ng et al., 2021). We will provide instructors with slides, speaker notes, suggested readings, discussion guides, and videos to help them and their students learn basic content about AI and effective practices for using AI for writing. Our instructional resources support instructors in making AI literacy

transparent and explicit as students engage with PapyrusAI and build strong writing skills for both independent and AI-assisted writing.

Conditions for Desired Change

Conditions required to support student improvement are access to PapyrusAI, a knowledgeable teacher, and instructional resources. For PapyrusAI access, students and instructors will require an internet-enabled device (e.g., Chromebook, laptop, or desktop computer), internet access, and the web-based application itself. Our community college partners will provide access to devices (if needed), internet, and technology support through existing programs and staffing. For effective use of PapyrusAI, students will benefit from instructors who have received PD on use of the tool and AI literacy, and support materials that help students progress from sign on through interactions with the AI. As part of this project, we will iterate the necessary instructional resources, including PD, and develop an Implementation Support Guide. Instructors will be taught effective methods for determining if and when AI tools would support both writing and AI literacy learning objectives, but will not be required to use them in any particular manner nor in any particular amount beyond a request to allow some form of use in a majority of writing assignments (i.e., instructors determine the appropriate types and frequency of use, quantity of use, and whether such use occurs in class or as part of out-of-class instruction).

Improvement Over Current Tools

Technology Solutions

Because of the centrality of writing to learning (Graham, 2022), as well as the difficulty of providing consistent feedback on large numbers of student writing samples, the development of automated writing evaluation (“AWE”) software has been a major focus of educational technology for decades (for an overview, see Ifenthaler, 2022; Shermis & Burstein, 2013). While some AWE systems are as efficient and reliable as human raters, they are typically less accurate, more generic and verbose, and confusing to feedback recipients (Grimes & Warschauer, 2010; Shermis & Wilson, 2024; Wilson & MacArthur, 2024). Additionally, preparing such tools for educational settings takes substantial time (Chen et al., 2023; Warschauer & Ware, 2006). Historically, AWE systems have required training on hundreds of essays written to the same prompt and calibration with human-provided feedback. These requirements increase the cost and reduce instructors’ flexibility in using these systems because they are limited to the specific prompts used for training. For these reasons, existing products (pre-LLMs) provide little benefit to student writing outcomes and have not achieved broad use in postsecondary institutions.

The most commonly used AWE products provide even less support to help students *plan* their writing. These products typically offer general rubrics and strategy suggestions, but do not attempt to dialogue with students about their writing plans as a skilled writing coach might do. Niche products, such as the IES-funded *WritingPal*, provide more adaptive and personalized tutoring on writing strategies but were developed with earlier forms of artificial intelligence and lack the ability to give detailed adaptive guidance to individual students on their own writing plans and were not successfully brought to market.

Over the last two years, there has been a great deal of attention to the role of generative AI in supporting student writing—and the development of students as writers—in postsecondary instruction. Teachers and learners across the country are making use of general purpose generative AI tools, such as ChatGPT and Microsoft Co-Pilot, for this purpose. But integrating these tools into writing instructions is challenging for a number of reasons, including the necessity for (and the cost of) personal subscriptions, threats to privacy of having data retained

for training purposes, lack of specifically developed research-based scaffolds, lack of instructor oversight to how tools are used, and lack of controls for discouraging misuse.

In response to these limitations, a number of generative AI tools have been developed with more specific pedagogical writing purposes in mind. Those appropriate for postsecondary use include Grammarly Go, MyEssayGrader, and CoGrader. However, each of these tools is limited to providing feedback on student writing; none of them also help students plan their writing—a key lever of learning to write well—nor do they actively promote teacher and student AI literacy, or allow teachers control over program prompts and modules.

PapyrusAI improves over existing tools in the following unique ways:

- It focuses not only on helping students *revise* their work, but also *strategize their writing plans*—a key element of learning to write well;
- It is designed to foster students' *AI literacy* by engaging them in direct back-and-forth interaction with the AI tutor, thus providing training wheels so that students can eventually transition to productive and responsible use of general-purpose AI tools.
- It leverages instructor expertise by allowing them to not only select among existing prompts, but also modify or create new prompts reflecting their teaching contexts—thus giving postsecondary instructors the kind of control and agency to make the platform maximally effective for their own classes.

Educational Practices

The most common approaches to supporting students on planning and revising their writing involve writing organizers, mind maps, and feedback from instructors and peers. PapyrusAI is an improvement because it can adaptively engage in dialogue with students to plan their writing, use open-ended questions to encourage students to expand on their text, provide specific personalized feedback, and parse the process of writing into manageable parts. PapyrusAI is available 24/7 and provides immediate responses, unlike instructors and peers who have limited availability. PapyrusAI also has unlimited patience, allowing students to question and reiterate as often as needed, with no social consequences.

Evidence of Need and Likely Use

The teaching and learning of writing have long been an area of concern, but instructors are challenged to find sufficient time and resources to meet this need (Graham, 2019). The existence of AI, which can provide planning support, feedback, and personalized coaching to students on any text, could free up instructors to provide targeted instruction and could also make formative assessment information accessible to instructors. However, instructors and institutions are grappling with when, if, and how to incorporate generative AI into writing without subverting the learning of necessary skills.

AI literacy is a newer, but critical, concern (Heikkila, 2023). The PIs have been inundated with interest in this project and support for development efforts as institutions grapple with how to prepare instructors, and instructors on how to prepare students, for this rapidly changing environment. Educators are anxious to find an approach to deal with the rise and diffusion of ChatGPT and similar AI tools.

Potential Market

Our target market is the over 1,000 community colleges in the United States and their approximately 9 million students (Fink, n.d.). The education technology market is expected to grow from \$137 billion in 2023 to \$345 billion in 2030 (Virtue Market Research, 2024), with

North America being the biggest contributor to this growth. Products leveraging AI will form a large part of this growth (MarketResearch.biz, 2023). We anticipate a growing demand for products that help students with academic skills while also improving their AI literacy.

Potential for Commercialization and Resources for Wider Adoption

Though there is great interest and need for AI-based postsecondary writing platforms, there is no product yet with a sizable market share. PapyrusAI was selected through a campus-wide competitive review as the single AI-based learning platform under development at UCI that has the greatest potential for commercialization. As a result, UCI Beall Applied Innovation—the main incubator for commercialization of UCI-developed knowledge—provided funding for the initial adaptation of the NSF-funded tool from their competitive “Proof-of-Product” commercialization program, which helps faculty researchers to advance their discoveries towards the market. As part of that award, our team participated in I-Corps™, an NSF-funded grant program, in which a team of experts train innovators to broaden the impact of research projects and develop effective solutions to customer problems. The I-Corps™ curriculum is based on real-world, hands-on learning that involves finding customers and understanding their problems to achieve a product/market fit. Most recently, Learn Capital—one of the oldest and largest ed tech oriented venture capital firms in the country—invited the PapyrusAI team to present to its annual investor meeting in March 2025. UCI Beall Applied Innovation remains ready to provide whatever resources are needed to commercialize for large scale up, including experts in sales, marketing, and customer service.

Overview of Center Plans

The WRITE AI Center will focus on the problem of practice of providing sufficient personalized writing instruction and scaffolded support to ensure postsecondary students gain the necessary writing skills to succeed in the workforce.

Exploratory Study: Ongoing national survey of developers and institutions on tools used in postsecondary writing instruction, culminating in a searchable database available to the public with key information on the top 100 tools in use. In-depth interviews with faculty and college administrators will further explore different use cases, illustrate challenges, and elevate lessons from promising practices.

Generative AI Tool for Use in Postsecondary Instruction:

- *Iterative development* of PapyrusAI with 24 community college instructors from 6 community colleges in Years 1 and 2,
- *Pilot of promise study: Implementation* through RCT with 60 instructors from the same colleges in Years 3-4 (measuring student writing outcomes and persistence, and student and faculty self-efficacy and AI literacy), and *cost analysis study*.
- Development of an *Implementation Support Guide* in Years 2-5.

Research Training: The WRITE AI Center will develop an online research hub to support both comprehensive learning through an annual fellows program and just-in-time access for the broader research community. Training programs will focus on:

- developing robust frameworks for evaluating the instructional effectiveness, scalability, biases, and ethical implications of generative AI tools;
- accessing, cleaning, and preparing tool data with attention to privacy and quality;
- measuring learning engagement and tool behavior using advanced data analytics;
- conducting causal research via A/B testing and other innovative methodologies; and

- employing mixed-methods approaches that integrate advanced statistical and qualitative techniques to understand generative AI tool usage and impact.

National Leadership, Capacity Building, and Outreach Activities: To guide policymakers, educational leaders, tool developers, educators, and the broader public in adopting research-based approaches to the use of generative AI in postsecondary education, the Center will:

- disseminate actionable information and research findings through a website, webinars, training programs, social media, research briefs, preprints, publications, and presentations;
- foster collaboration by convening stakeholders in an annual summit, innovation labs, and national conferences; and
- create and share resources to support effective use of generative AI in postsecondary education.

Key Partners and Research Sites: UCI, MDRC, American Council on Education (ACE), California Community Colleges (CCCs), the Virginia Community College System (VCCS), and Houston Community College (HCC) are collaborating to combine their expertise on writing, generative AI, development, research, and postsecondary education to benefit the WRITE AI Center. UCI brings deep expertise in postsecondary education, the research of writing, and the use of digital tools for literacy, including the PapyrusAI tool developed internally. ACE is committed to coordinating policy and leadership work on improving higher education among its over 1,600 member institutions. MDRC brings a strong track record in research and evaluation related to applications of artificial intelligence, as well as an extensive track record in postsecondary education. The CCCs are represented by Foothill DeAnza (FHDA) Community College District—a large Silicon Valley-based district which leads a number of relevant systemwide digital learning initiatives, including the California Virtual Campus (CVC), the Online Network of Educators (@ONE), and a new AI-focused digital center for innovation and transformation (Digital Center); FHDA will also coordinate Fullerton College, Saddleback College, and Reedley College, the three CCCs that will be our partners in the development and implementation work. VCCS has been a national leader in educational reform with technology, and will coordinate efforts with Northern Virginia and Virginia Western Community Colleges for our development and implementation work. HCC focuses on providing a high-quality, affordable education for academic advancement, workforce training, career development and lifelong learning and, as a major urban campus in Texas, adds geographic diversity as an additional site of the development and implementation study. A group of seven advisors will add complementary expertise in postsecondary writing instruction, AI-based instruction, and the ethics of generative AI use.

Our aims and research questions for the WRITE AI Center’s activities are: **Aim 1: To develop and make available information about generative AI tools used in postsecondary writing instruction.** What are the most widely used generative AI tools for teaching and learning of postsecondary writing and what are their relevant features for instructional adoption and use?

Aim 2: To develop and innovate PapyrusAI, professional development, and related instructional resources (collectively, the “intervention”) so that community college instructors can use AI to improve students’ writing and AI literacy.

Aim 3: To examine the usability and feasibility of the intervention in community college writing instruction. (a) How much do instructors and students use PapyrusAI?; and (b) Is it feasible to implement PapyrusAI in the writing curriculum?

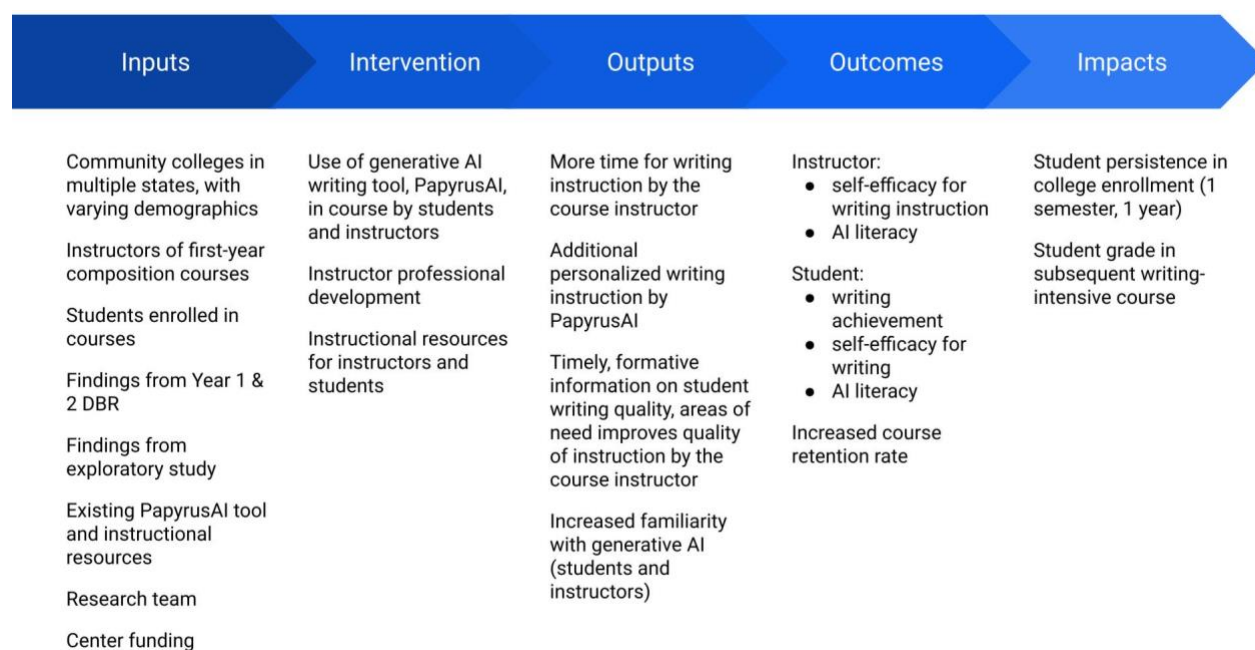
Aim 4: To evaluate the potential promise of using PapyrusAI on student writing and AI

literacy outcomes and explore potential moderators. (a) Do instructors and students report more time for writing instruction by the course instructor?; (b) Do students report receiving additional personalized writing instruction by the tool?; (c) Do instructors and students report that having timely formative information on student writing quality and areas of need improves the quality of instruction?; (d) Do instructors and students have an increased familiarity with generative AI?; (e) How does the intervention impact instructors' and students' self-efficacy? (f) How does the intervention impact instructors' and students' AI literacy? (g) How does the intervention impact student writing quality?; (h) How does the intervention impact course retention rate?; (i) How does the intervention impact student persistence in college enrollment?; (j) How does the intervention impact students' grade in subsequent writing-intensive courses?; (k) Are there heterogeneous effects for students who have differences in their baseline writing or AI literacy skills?; and (l) Was the intervention implemented with fidelity?

Aim 5: To estimate the cost of implementing the intervention. What is the cost of implementing PapyrusAI?

Aim 6: To develop an Implementation Support Guide. What do researchers and practitioners need to implement the tool in community college writing instruction?

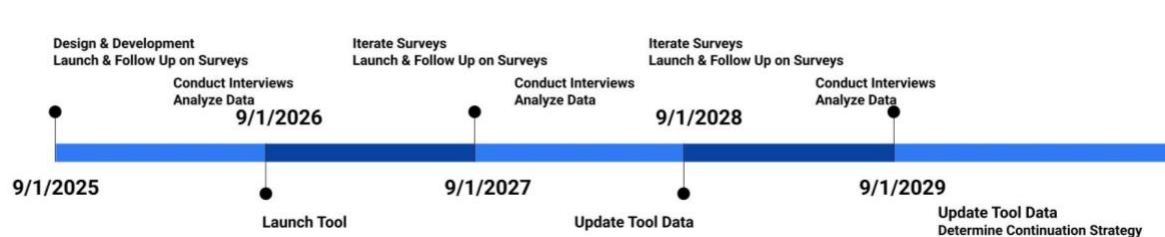
The **timeline** for our full proposal is in Appendix B-1. Our RCT will be based on the **Theory of Change** in Appendix B-2 and below.



Research Plan for the Focused Program of Research

The Center will carry out a focused program of research on the use of generative AI in improving *writing* instruction in *postsecondary education*. The focused program of research will include (a) an exploratory study that will identify the current use of generative AI in *postsecondary writing instruction* and (b) iterative development and piloting of *PapyrusAI* and related instructional resources for *writing instruction in community colleges*.

Exploratory Study



Identify Current Use of Generative AI in Postsecondary Instruction

The Center will identify generative AI tools being used to improve postsecondary writing instruction. This work will be ongoing as long as the Center exists, and the findings will be provided and updated on the Center’s website in a publicly accessible and usable reference for researchers and practitioners.

This study will address **Aim 1: To develop and make available information about generative AI tools used in postsecondary writing instruction** and the related research question, What are the most widely used generative AI tools for teaching and learning of postsecondary writing and what are their relevant features for instructional adoption and use?

The exploratory study will identify generative AI tools that writing faculty are using to improve writing instruction at public community colleges and universities across the United States, and it will describe and catalog the features of these tools for a non-technical audience. The study will focus on identifying AI tools used in the core writing courses these students take, such as first-year composition and developmental writing, as well as the subsequent disciplinary writing courses like Business Writing.

The study will collect information about the AI tools being used broadly from providers and institutions across the country and will dig deeper into institutional practices to draw out key examples and lessons. The team will: (1) identify potential information sources about AI tools using a combination of two main starting points: a comprehensive set of providers of generative AI tools that can be identified through public data sources; and a representative sample of public colleges and universities who are using those tools; (2) administer two survey questionnaires, one for AI tool providers and one for colleges and universities; and (3) conduct in-depth interviews with key staff and instructors at selected institutions to more deeply explore their use of the tools and their perspectives on how the tools are helping to improve instruction.

A major objective of the exploratory study will be to learn about the full set of AI tools being used to improve college writing instruction across the United States. To identify these AI tool providers, the team will first start with web searches for available generative AI tools, publicly available lists of tools, and the team’s knowledge and relationships with existing providers. The survey questionnaire for the AI tool providers will focus on characteristics of the tools themselves, such as: guides and manuals, costs, infrastructure requirements, training needs; how the tool addresses safety, bias, and fairness; and available evidence of each tool’s impacts on postsecondary students’ academic outcomes. In many cases, information needed to fill out the questionnaire will be available in publicly available websites and documents about the tools. To fill in any remaining gaps in the information, the team will follow-up with the providers by email and phone with requests for the remaining specifics.

Information collected from these providers will also include identification of the colleges and universities that are using the AI tool. The team will then contact those institutions to collect additional information using a survey questionnaire focused on learning about which AI tools they are using, how they are applying the tools to support instruction, and in which writing courses they are using the tools. The survey questionnaire will be designed to be flexible so that

multiple individuals at each institution could respond if more than one tool is in use in different classrooms. The team will also pilot a process to identify the correct institutional respondents, knowing that the relevant information may reside with different people at different institutions, such as department chairs, writing centers, or faculty, and that institutional leadership may need to be engaged to correctly identify the full set of respondents and encourage participation.

A second major objective of the exploratory study will be to understand and document the landscape of institutions and instructors who are using generative AI tools in writing instruction, and to explore patterns that arise within it. To achieve this goal, the team will draw a random sample of colleges and universities chosen from among all public institutions, and will administer the institution-specific survey questionnaire to them, including questions about which tools instructors are using. In many cases these institutions will be using tools for which the team previously collected information as described above. In some cases however, this process may uncover new tools. In those cases, the team will reach out to those tool providers to collect additional information about the tools using the provider-specific survey questionnaire.

The team will also conduct longer interviews with key staff and instructors at selected institutions to better understand their perspectives and how they are using the generative AI tools. The goal will be to dig deeper into use cases, explore promising practices, potential pitfalls or challenges, draw out lessons, and highlight other examples to inform the field.

The data collected from the survey questionnaires will be coded and compiled into a publicly available and searchable database with a web-friendly user interface to help practitioners, policymakers, and other researchers search among the tools. The study team will analyze the data to explore patterns in which colleges are or are not using generative AI tools to improve writing instruction, as well as patterns in which tools they are using and how they are using them. Findings from these analyses will be included on the website, alongside findings from the qualitative interviews which will highlight important use cases and examples. The full set of data collected from the survey questionnaires will also be downloadable for other researchers to analyze. The information collected will also be shared with the team creating our Implementation Support Guide, discussed below, to help guide their understanding of the necessary expertise, conditions, and resources for successful implementation of such tools generally and the development of assessments to check implementation readiness.

The website will be regularly updated with new data and analyses. The surveys will be administered in three waves over the course of the Center, with new data posted at the beginning of Year 2, at the end of Year 3, and during Year 5. Analyses and posted findings will be ongoing throughout the life of the Center.

We will also work with the Center's Advisory Board and ACE's membership to solicit feedback about the information collected and how it is presented to ensure its usefulness for the field. For example, in Year 1, draft survey questionnaires will be shared with advisors for feedback to ensure that the most important information is collected up front. Early prototypes of the website will also be shared to ensure that the information is easily accessed and navigable by diverse audiences. Advisor feedback will continue to be important in later years, and ACE will also solicit feedback from its membership through its normal course of member communications and through dissemination activities associated with the Center and the exploratory data set. This later feedback will inform not only the presentation of the data but also ongoing data collection activities such as the focal areas for qualitative interviews.

Generative AI Tool for Use in Postsecondary Instruction

PapyrusAI and the related instructional resources will be further developed, piloted, and improved for use in community college first-year composition courses. At all stages of the process, the research team will work closely with community college instructors and students to ensure that humans stay at the center of everything we do together. Iterative design and testing in these real-world contexts ensure the tool is responsive to the unique needs of community colleges, including accessibility for broad student populations, usability for instructors with varying levels of technological proficiency and resources, and alignment with institutional priorities and measurable student success. The combination of our iterative design and development, partnerships, and focus on sustained change through implementation aligns with principles of DBIR (Fishman et al., 2013). As such, our research on using PapyrusAI in postsecondary education will draw from a variety of methods and data sources, described in the activities below to address **Aim 2: To develop and innovate PapyrusAI, professional development, and related instructional resources** so that community college instructors can use AI to improve students' writing and AI literacy in first-year composition courses. Concurrently, we will address **Aim 3: To examine the usability and feasibility of the intervention in community college writing instruction**, and the related research questions: (a) How much do instructors and students use PapyrusAI?; and (b) Is it feasible to implement PapyrusAI in the writing curriculum?

Setting

Study 2 will be situated within the context of community colleges. We will specifically work with 3 community colleges in California, 2 in Virginia, and 1 in Texas (see Appendix B-3 for demographic information on these partners). Our college partners were chosen across settings to encompass urban, suburban, and rural settings across the nation. It is our intention to work in partnership with these 6 colleges to ensure that our research is applicable to different systems, geographical locations, college sizes, and demographic contexts to increase the generalizability of our findings and usefulness of our tool.

The main locus of our development work (and the sole focus of our pilot study, see below) is **first-year composition**. First-year composition is one of the most widely taught undergraduate courses in the US, and an almost universal requirement for graduation. Within community colleges, it is the single most widely taught course as well as the main impediment toward higher education success, with non completers of the first year college composition course 20 times as likely to drop out than completers of the course (Community College Research Center, 2013). The size of this course and its critical role as a gateway to further college and career success motivate our focus on it. The course is a semester long, so each year our participating instructors will generally teach two classes total (Virginia community colleges have a reduced offering in spring, so those instructors may not all teach a second course). The fact that instructors will teach the course more than once using the intervention will allow us to observe changes as instructors' familiarity with the intervention increases and the implementation matures.

We will place secondary focus during the development phase on two other sets of courses, those in **developmental writing** and **disciplinary writing**. Developmental writing courses are designed to help students who are not yet prepared for college-level writing courses and need instruction in basic skills such as sentence structure, paragraph development, organization, grammar, and mechanics. Disciplinary writing courses teach students the particular conventions,

language, structure, and forms of rhetoric required to communicate effectively within a specific academic or professional community. Among disciplinary courses, we will focus specifically on business writing, which is the most widely taught disciplinary writing course at our partner community colleges and one that has an especially interesting relationship to AI-based writing, given the rapid growth of AI use in business communications (see, e.g., Hawkins, 2025).

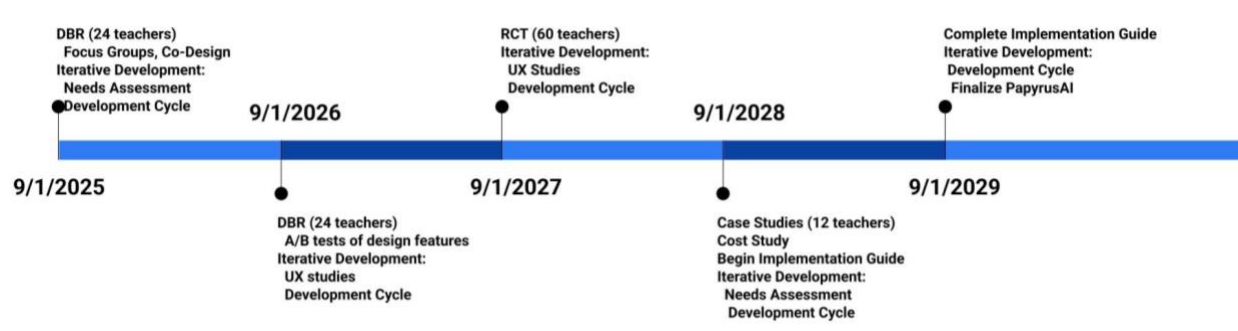
We situate our work in first-year composition and surrounding courses, because we believe that it is a leading edge case for learning how to productively and responsibly navigate the tensions created by generative AI. In addition, because most students take first-year composition, educating students on the proper use of generative AI allows institutions to ensure that the majority of students are AI literate as they progress in both college and career.

Recruitment

In Years 1 and 2, we will work closely with 24 instructors, with at least 2 first-year composition instructors at each of our 6 institutions, and with developmental and disciplinary literacy instructors across the colleges comprising the remaining 12 instructors. We will be able to work with instructors over 4 semesters to iteratively develop, implement, and revise the intervention so that by the end of Year 2 our implementations will be fairly mature. Based on information provided by the colleges, we estimate that instructors in the study will teach an average of 20 students per class each semester, for a total of 40 students per teacher annually (960/year). These instructors will be identified by their college as expressing strong interest in using AI and working with our research team. For the Year 3 RCT, we will offer participation to 60 first-year composition instructors, 10 at each college. In order to understand the practices of exemplary instructors, in Year 4 we will study 12 case study instructors using qualitative methods. The 12 instructors will be invited based on their students having shown the greatest gains in writing improvement and/or AI literacy during the RCT and balanced to ensure coverage of different contexts. Control instructors will access the intervention in Year 4. Over the course of the project, we will offer the intervention to 84 unique instructors and 3,600 unique students. See Appendix B-4 for a table of participating instructors and students.

Strategies to Increase Participation and Reduce Attrition

With respect to participation, we believe that the compensation provided to instructors (see Budget) and their pressing need for assistance in addressing the challenges of AI will make our recruitment goals very reasonable. Based on our prior work in community colleges, we anticipate 10% attrition for instructors and 15% for students. To reduce instructor attrition, we will keep clear and consistent communications with participating instructors throughout the project: We will ensure regular check-ins and open channels of communication to address any concerns or challenges faced by the instructors and adopt a solutions-oriented stance.



Development of a Generative AI Tool: Design-Based Research

The development under this grant will be iterative and guided by the findings of our design-based research (DBR; Anderson & Shattuck, 2012; The Design-Based Research Collective, 2003) with instructors and students. Our iterative development approach is informed by DBR, agile software development workflow, participatory design, and the Plan-Do-Study-Act framework. We have found that these complementary and often overlapping approaches lead to successful development of curricular interventions that include technology design. Ultimately, we will use an iterative design process with cycles of requirement analysis and design followed by development. This allows the development to be flexible and responsive to the results of the co-design, to ensure that the tool best complements the instructional resources and meets the participants' needs, and the professional development provides the appropriate support for proper use of both. Our research group has substantial experience conducting participatory design for and with instructors and students, including the initial design of the intervention in upper division disciplinary writing at UCI.

In Year 1, participating instructors will attend two 90-minute PD sessions on the basic information about AI, how to use PapyrusAI and the instructional resources created for the original university implementation, and how to incorporate AI into their writing instruction. They will practice using the tool, become familiar with it, and engage in a collaborative discussion of the specific needs and resources their students currently have. Researchers and instructors will meet monthly to discuss implementation challenges, collaborate, provide feedback on the tool and materials, and receive targeted PD in response to identified needs.

During the year, we will work with these instructors to refine critical core components of the tool, including best uses of the tool for supporting student writing, aligning the capabilities of PapyrusAI with the learning objectives of the course, and understanding the AI literacy information and skills students need to effectively (and safely) interact with the tool. We will specifically discuss how the user interface of the tool can be improved, what revisions and additions to the prompt library are appropriate, and determine the types of instructional resources needed to support instructors and students. We will use this iterative process to identify, evaluate, and collaboratively determine what features of the tool and instructional resources are most pedagogically appropriate and weigh that against the development time and expense for implementation. We will also discuss the findings of the exploratory study to determine what features and functionality identified in the available writing tools might be valued in our context. Through cycles of needs assessment, UX studies, and development (see Appendix B-5 for the development timeline), we will develop and refine the intervention by explicitly and systematically adapting the PapyrusAI tool, the embedded prompts, and our instructional resources to support student writing and AI literacy and integration into the existing curriculum.

In Year 2, the same 24 instructors will continue to integrate the tool and materials into their courses, along with participating in A/B tests to assess various design features and options, such as whether the information on the teacher dashboards provide relevant and actionable information clearly, whether the feedback from different versions of our prompts is developmentally appropriate, and other decisions requiring testing prior to the larger implementation in Year 3. In addition to this iterative development work, we will assess usability and feasibility of the intervention. For this, we will gather data from the tool indicating the time, type, and amount of use by instructors and students; conduct interviews and observations; and collect teaching and learning artifacts indicating which assignments use the tool and resources and student writing for assignments related to tool use. **Usability** refers to the ease of instructors and students to access, log into, navigate, and use the output from PapyrusAI and related

resources. We will similarly assess *feasibility* by analyzing the degree to which instructors incorporate PapyrusAI into their curriculum. These data will be analyzed both in the aggregate and also separately by characteristics to understand differences between contexts that might be addressed through additional instructional resources or design changes.

Improvements to PapyrusAI

In addition to implementing changes inspired by our DBR, we intend to improve the output of PapyrusAI by implementing a multi-agent system (the current tool uses a single generative AI model). Recently, based on the development of using one LLM as a single planning or decision-making agent, LLM-based multi-agent systems have achieved considerable progress in complex problem-solving (Guo et al., 2024). Compared to systems using a single LLM-powered agent, multi-agent systems offer advanced capabilities by specializing LLMs into various distinct agents, each with different capabilities, and enabling dynamic interactions among these diverse agents to effectively engage in planning, discussions, and decision-making (Guo et al., 2024; Han et al., 2024). We believe that a multi-agent system will be able to provide more guardrails to ensure the use of the tool is pedagogically sound and increased information to instructors on students' AI use and writing quality.

Instructors have asked for increased certainty that the tool will not write for the students as well as analytics that will help them provide more just-in-time, personalized instruction to their students. We also want to make sure that the tool is coaching the student in research-based ways that take into account best practices for effective feedback, retaining student motivation and agency. To address these desires, we will create additional agents that automatically monitor students' conversations with PapyrusAI for different purposes. For example, a *Teaching Coach* could oversee the conversation between the AI and student to make sure that the tool is being educational, using research-based pedagogy, and not writing on the students' behalf. The *Teaching Coach* would be able to inject hidden messages into the conversation when necessary to guide the interaction in productive ways. Another agent might be a *Learning Analyzer*, which would identify what elements of the writing process the student may be struggling with so that the tool could provide the instructor with actionable information on the needs of the students in their class. We will also be able to fine-tune a model (or implement a RAG-based agent) on the student writing data using the holistically scored writing assessments from the first years of the project and pilot a scoring feature in the instructor dashboards in Year 3 if the instructors believe that would be of value and is sufficiently accurate. With respect to the information provided to instructors by the *Learning Analyzer*, we will work with our instructors to determine what instructional data is useful and how to best present it as actionable information.

Improvements to the Prompts and Instructional Resources

Our prompt design work will focus on creating new and refining existing prompts based on evolving knowledge of prompt engineering techniques, pedagogical needs in context, and tool and LLM changes (e.g., to take advantage of multi-agent ability). As part of our development work, we will incorporate strategies for planning, drafting, and revising, as well as for goal setting, task management, progress monitoring, and reflection found effective by advisor MacArthur (MacArthur et al., 2015) for self-regulated strategy instruction in college developmental writing. Our instructional resources will be similarly revised to provide support for these strategies at the instructor and student levels. These strategies give students a systematic way to approach the complex task of writing and reduce the cognitive load. They also improve student self-efficacy, which in turn has been shown to improve writing quality (Bruning

& Kaufman, 2016; Pajares & Valiante, 1997).

Our researchers, who have to date created over 100 prompts for writing with AI, will take the lead on prompt design and testing, bringing any decision points (e.g., how should the tool describe the role of a thesis statement) to the instructors for input on the best instructional choice. Instructors will provide guidance on the types of AI output and student-AI activities that would be most developmentally appropriate and consistent with the existing curriculum across our systems, support students coming from developmental courses, and provide students with the appropriate vocabulary and skills to support disciplinary literacy courses in the future. See Appendix C-1 for examples of our extensive prompt development and testing.

We will monitor student-AI interactions and gather instructor and student feedback during interviews and focus groups to ensure that the new feature(s) and prompts do not reduce the usability of PapyrusAI. The researchers will conduct qualitative analyses of the student/AI interactions to determine whether the feedback and interactions improve in quality as well as ensure that the instructors help guide any necessary adjustments.

Instructional materials for students will be implemented in classes while instructors and researchers observe how students respond. If instructors or researchers suggest re-design or supplemental materials that are aligned with our learning objectives, we will collaborate to make the necessary changes or products for the next phase. PD and resources to support instructors' implementation will be improved from the existing materials and refined to meet the needs of the instructors and students in our context.

Decision Points

Our decision points will be in each of three areas: tool design, prompt design, and instructional resource design. We expect our tool design work will focus on improving user interfaces; implementing increased functionality (particularly implementation of the multi-agent system); and taking advantage of improved commercial models. Our development plan allows us to plug in new large language models with relative ease. We have already done this twice in the first 18 months, from GPT-3.5 to GPT-4 to GPT4-o. Changes are implemented in a test environment and rolled out when they will not conflict with classroom activities (e.g., during winter and spring breaks). Co-I Lopes will be particularly helpful in providing computer science expertise as we make these decisions with respect to tool design, while AB members MacArthur and Graham will support decisions on prompting and instructional materials.

Pilot of Promise Study: Implementation

In Year 3, we will scale up to an RCT in **first-year composition courses** at our 6 colleges with 60 instructors, randomly assigned to treatment or control, to examine the potential promise of PapyrusAI on teacher and student outcomes. The RCT addresses **Aim 4: To evaluate the potential promise of using PapyrusAI on student writing and AI literacy outcomes and explore potential moderators**. For the associated research questions, see Overview of Center Plans above and Appendix B-6.

The project includes a number of design elements and procedural safeguards to promote high internal validity and the education community's confidence in results. First, we utilize a rigorous RCT to minimize alternative explanations for findings (Shadish et al., 2002). We will employ a high-quality design aligned with the Standards for Excellence in Education Research (SEER) by clearly identifying the core components of our intervention (tool, embedded prompts, and instructional resources) and focusing on meaningful student and teacher outcomes, using technically valid measures of outcomes, randomly assigning teachers to RCT conditions, clearly

defining our sample, documenting and responding to attrition rates, documenting our procedures (and the counterfactual) for replication, and analyzing the total costs of our intervention. MDRC will select an expert uninvolved in recruitment, data collection, and development to randomly assign participating instructors within a college to conditions to minimize selection bias and other potential threats to the validity of causal inferences. MDRC will also review procedures and ensure they are sufficiently specific to ensure the integrity of the research. Treatment instructors will attend the PD and utilize PapyrusAI and the instructional resources in their classes. Counterfactual instructors will conduct their classes as usual, with no implementation of the intervention, although they may use other generative AI in their classes to the extent desired. Those conducting student assessments will be blind to students' assigned condition.

We will also take additional precautions and strategies to prevent threats to validity, including (1) clear communication to colleges, instructors, and administrators about the study design and the importance of maintaining its integrity; (2) strict adherence to randomization; (3) conducting monitoring via regular check-ins and regular on-site visits to ensure proper adherence to study protocols; (4) careful documentation of implementation fidelity (see below); (5) prespecification and open sourcing of analysis plans; and (6) conducting sensitivity analyses to assess the robustness of findings and evaluate potential impact of contamination. Second, we will minimize attrition using procedures and strategies that were highly successful in past work (see above). As with all projects, some missing data is inevitable and will be addressed via both maximum likelihood and multiple imputation approaches or missing not at random (MNAR) methods (Baraldi & Enders, 2010). We will make findings and data publicly available as discussed below.

The counterfactual condition will be business as usual (BAU), enabling us to infer the effect of PapyrusAI compared to current practices in community colleges. This means that our findings will be directly applicable to real-world contexts, as BAU accurately reflects the current practices. Moreover, including a BAU condition enhances the generalizability of project findings to broader settings. In addition, in line with SEER principles, we will carefully observe and document instructional practices in the counterfactual BAU condition.

Fidelity will be measured as an implementation outcome. We adhere to a broad conceptualization of fidelity commensurate with the literature, comprising adherence/responsiveness, quality, dosage, and differentiation (Capin et al., 2018; Hill & Erickson, 2019). Specifically, we will measure *adherence* (% of instructors and students using PapyrusAI); perceived *quality*—along with feasibility and acceptability—of PapyrusAI and the instructional resources (including PD attendance) via the adapted Perceived Characteristics of Intervention Scale (Cook et al., 2015) and the Usage Rating Profile–Intervention (Briesch et al., 2013; Chafouleas et al., 2009), along with open-ended items on teacher surveys; and *dosage* as the use of PapyrusAI by instructors and students. We will measure *differentiation* by comparing conditions on baseline student, teacher, and college characteristics and using classroom observations to document instructional practices in both PapyrusAI and BAU comparison conditions (see additional details below). If we notice that PapyrusAI activities are not being implemented or only implemented to a limited extent, we will meet with those instructors to discuss barriers, facilitators, and strategies to meet implementation goals. We anticipate strong, collective buy-in, and thus fidelity, given our community college partners' interest in PapyrusAI. Instructors will receive incentives (see Budget) for research-related activities.

Implementation fidelity will be measured using tool data to analyze the interactions between students and the tool, as well as the number and duration of interactions with

PapyrusAI. Implementation fidelity will also be measured through classroom observations, teacher interviews, and student focus groups. We will collect artifacts from instructors including lesson plans, assignments, and writing samples, which will provide additional evidence of fidelity of implementation (e.g., was the tool permitted in a majority of assignments?). This evidence will also be used to understand local adaptations and necessary conditions for successful implementation for our guide discussed below. Completion of the PD will be documented by taking attendance and noting any early departures.

Documentation of the counterfactual/BAU condition will include interviewing and observing control instructors to gather information about any use of AI in classes *not* participating in the intervention.

Year 4 will consist of 12 qualitative case studies of exemplary instructors to provide context and texture to our findings and finalization of our intervention for public dissemination. Our control instructors will also be provided access to PapyrusAI and the instructional resources in Year 4 (a control-waitlist design).

Measures

A table of the alignment between the measures and the research questions is in Appendix B-6.

Qualitative

Teaching and Learning Artifacts (Aim 2, Aim 3, Aim 4). Instructors' syllabi and course materials will be collected to understand how PapyrusAI and instructional resources were implemented in the course. We will also be able to gauge the amount and type of writing students are tasked with over the year. Students' texts and coursework will be collected and analyzed to shed light on learning and writing processes, and the quality of writing and AI literacy. Instructor evaluations of student work will also be collected and analyzed. We will be able to compare differences between treatment and control classes in the RCT.

PapyrusAI Data (Aim 2, Aim 3, Aim 4, Aim 5). We will use clickstream data from the tool to analyze student input and AI output, including personalized instruction. Analysis of this data will include qualitative coding of student and AI conversations, and use of natural language process tools (e.g., automated topic and sentiment analysis).

Classroom Observations (Aim 2, Aim 3, Aim 4). First-year composition classes will be observed monthly, with structured observation notes taken in all observations. We will initially use a protocol that focuses on co-design issues raised by the group, how the instructors are implementing the tool and resources, how students appear to be developing as writers and their AI literacy, how the tool is being used, and whether the intervention appears to be effective. Our decisions and choices will be carefully documented. The iterative nature of the intervention will allow us to address the tension between adaptation and strict fidelity as the intervention is designed and implemented. During the RCT, we will randomly select 12 treatment and 12 control classes to observe (2 from each community college partner). Our observations will shift to prioritize an understanding of fidelity of implementation, local adaptations of the intervention, and BAU conditions to inform our analysis of moderators, a description of how the intervention may be impacting classroom practices, and the revision of the Implementation Support Guide as discussed below.

Teacher Interviews (Aim 2, Aim 3, Aim 4). We will interview instructors each semester to ask them about a range of topics covering tool design, instructional practices, and the impact of the intervention on their teaching and students' writing and AI literacy development. Questions will

include the ways in which they incorporated PapyrusAI into their course, how the formative data from PapyrusAI impacted their instruction, what suggestions they have for improving the teacher data and presentation of data, what they found most effective and most problematic in implementing the intervention, what they plan to do in future courses, any unique patterns of student use they notice, and what supports and scaffolding they think are needed to implement PapyrusAI effectively. We will note how the tool is being integrated into the procedures, pedagogical practices, digital learning tools, curricula, and assessments of our partners (and across differing colleges). We will also note all costs associated with adoption of the intervention. During the RCT, we will interview the instructors selected for observation. Interviews will prioritize an understanding of fidelity of implementation, local adaptations of the intervention, and BAU conditions.

Student Focus Groups (Aim 2, Aim 4). Students will be invited to participate in focus groups of 6-10 students each year to discuss their experiences in using PapyrusAI and the instructional resources as well as the impact the intervention had on the amount and quality of writing instruction they received from the instructor. These semi-structured discussions will focus on the user interface of the tool (in Years 1 and 2), what students found helpful, what challenges they faced, how they believe the use of AI writing tools impacted their development as writers, what supports they would have liked to have available, and suggestions for future improvements. We will include an intentional sample of students across the community colleges: proficient users of AI, reluctant users, and students at varying levels of achievement.

Team Meeting Notes, Conjecture Maps, and Other Artifacts From DBIR (Aim 2, Aim 3). We will take notes at all meetings of the research team, including meetings with the participating instructors. Notes, conjecture maps, and other artifacts will be coded to understand the themes, problems of practice, and opportunities arising. Conjecture maps (Sandoval, 2014) will iteratively be developed, shared, reviewed, and revised at all of these levels. At the beginning of the project we will develop conjectures about how the project is “embodied” (including the PapyrusAI tool; underlying prompts; student-facing learning resources such as necessary AI literacy materials, and teacher-facing learning resources including PD parameters; task structure; participant structure; and discursive practices); its mediating processes (how participants engage with those elements of embodiment to try to achieve intended outcomes); and our intended and expected outcomes. We will review these conjectures regularly.

Data Analysis Plans. Data analysis of the qualitative measures will take place on an ongoing basis to inform the iterative development of PapyrusAI, with data collection and analysis occurring simultaneously each year (Corbin & Strauss, 2008), and data from prior years used to inform subsequent years’ data collection (Coburn, 2004). These data will also directly inform the improvement process and the modifications made in the DBIR process, including adjustments to improve future scaling and integration into diverse settings, and our development of the Implementation Support Guide.

Coding will be based on the flexible coding methods of Deterding and Waters (2021). We will begin by indexing transcripts with broad codes that reflect the questions asked and the concepts under examination, then review the data for subsequently more fine-grained analyses, including analyses across respondent attributes and other heterogeneous factors. The index coding organizes information so that relevant sections can be looked at together, fostering increased consistency in coding (Katz & Hightower, 2023). Within the index codes, we will use analytic codes to identify emergent themes both within and across each index code. Working in

Dedoose, we will identify the relationships between the codes and look for heterogeneity among our instructors, courses, and student demographics. This flexible coding is optimized for qualitative data analysis software and large amounts of data. Flexible coding is also more reflective of studies, such as the current one, that are grounded in prior research and not purely inductive. In our case, we had an understanding of key issues typically involved in incorporating technology into classroom settings, for example (Warschauer, 2006), and the implementation of prior versions of the tool at UCI. We will also explore the use of generative AI and Natural Language Processing approaches to assist with coding, attending to current best practices as they develop (Morgan, 2023).

We will attend to disconfirming evidence of trends (Yin, 1994) and triangulate across the data (e.g., themes that emerge across observations, co-design groups, and artifacts have triangulated data). We will make extensive use of memo-writing to preserve and analyze information. For instance, as we code we will write memos that describe individual codes, overarching themes, or themes that seem to appear among distinct subgroups of respondents. These memos will form the basis for iterative design. Finally, we will conduct participant checks to ensure that we are appropriately interpreting their experiences.

Quantitative

Writing assessment (Aim 4). Writing quality will be assessed by source-based informational or argumentative writing tasks adapted from assessments in use by the participating teachers and scored by trained raters overseen by Co-I Hebert (the UCI team has repeatedly trained holistic scorers for this purpose during prior grants and achieves strong reliability consistently). Our writing assessment procedures align with SEER principles for using high-quality measures that are appropriate and for the context, valid and reliable (holistic quality has a strong track record in writing research), have practical significance for the instructors, and are not over-aligned with our PapyrusAI intervention. Working with first-year composition instructors in all 6 colleges, we will review existing course writing assignments and select 4 assignments in fall of Year 1. We will randomly assign each instructor in the study to have their students complete 1 of the 4 assignments during the semester. Researchers and instructors will review student writing samples to refine the assignments and align them across the colleges (to create a uniform assessment for this project), and select 2 of the assignments to give to students in the following semester, spring of Year 1. Writing sample scores will be shared with instructors, who will determine any additional instructions that should guide the raters to ensure that the holistic scores are aligned with instructors' practices and indicate any further revisions to the assignments. The revised writing assessments will be piloted in classrooms in Year 2 and final versions will serve as the measure in the RCT.

AI Literacy (Aim 4). Instructors and students in treatment and control classrooms will complete the GLAT: The Generative AI Literacy Assessment Test (Jin et al., 2024) at the beginning and end of the school year. This is a 20-question multiple choice test examining understanding of AI general concepts, logic systems, machine learning general concepts, and supervised learning. Structural validity and reliability were confirmed with responses from 355 higher education students using classical test theory and item response theory, resulting in a reliable 2-parameter logistic (2PL) model (Cronbach's $\alpha = 0.80$; omega total = 0.81) with a robust factor structure (RMSEA = 0.03; CFI = 0.97). Critically, GLAT scores were found to be significant predictors of learners' performance in GenAI-supported tasks, outperforming self-reported measures such as perceived ChatGPT proficiency and demonstrating external validity.

Self-Efficacy (Aim 4). We will measure instructor self-efficacy for teaching writing and student self-efficacy for writing using validated measures at the beginning and end of each semester. Instructor self-efficacy will be measured with an eight-item scale on efficacy for teaching writing taken from Graham et al. (2001) with items such as “I have effective ways to teach writing,” and “I can help students with the most difficult writing problems” (omega 0.78-0.91 across studies, Graham et al., 2023). For students, we will use the Self-Efficacy for Writing Scale (SEWS; Bruning et al., 2016) adapted to include revision (Chung et al., 2021), a measure we have been using in our current NSF study at UCI. The measure has questions on ideation, mechanics, self-regulation, and revision and had an alpha of 0.90.

Acceptance and Use of Generative AI (Aim 4). We will measure student and instructor change in technology acceptance and use of generative AI (which we refer to as “familiarity”) using the validated measure for generative AI acceptance of use (Strzelecki, 2024). The alpha for the various components of the measure range from 0.93 to 0.83. This measure is currently being used in our NSF study at UCI.

PapyrusAI Data (Aim 2, Aim 3, Aim 4, Aim 5). We will use clickstream data from the tool to analyze student input and AI output, including the number and duration of interactions. PapyrusAI data includes: time stamps and counts of all interactions (e.g., when students log on, how long they engage in interactions, how many times they interact); the number of tokens used in each interaction (pricing of the use of the API will be based on this); how many classes and modules a teacher creates; which prompts a teacher enables; and how often a teacher looks at student reports. Analysis of this data will include descriptive quantitative analysis of statistics related to usage count, duration, and time of day of interactions.

Institutional Data (Aim 4). We will gather data on students’ enrollment in the following semester and following year, as well as their grade in the subsequent writing intensive course. This will provide a distal measure in alignment with the SEER principles. We will also gather data on student demographics for heterogeneity analyses.

Data Analysis Plans. For the RCT, instructors within a college will be randomly placed into PapyrusAI and BAU conditions. We will compare quantitative student outcomes (pre/post writing quality, AI literacy) between these two conditions. We will monitor contamination through surveys, focus groups, and classroom observations and probe the robustness of our results to contamination using sensitivity analyses (e.g., Rhoads, 2011). Attrition across the RCT will be monitored via the number of students who withdraw from each condition. Based on prior research with similar populations, we anticipate attrition of approximately 15%. If differential attrition across groups is minimal, an intent-to-treat analysis (Ellenberg, 1996) will be used. In the event of differential attrition, we will assess baseline balance on observable characteristics between treatment and control groups among the remaining sample. If imbalances are detected, we will include relevant covariates in our models to adjust for potential confounding. We will also explore inverse probability weighting or doubly robust estimators to address potential selection bias due to attrition. To account for the primary clustering structure (students nested within instructor and college), a three-level model will be used and we will use clustered standard errors and fixed effects to account for state. We will examine the impact of assignment to PapyrusAI on student outcomes using a three-level set of equations that include a student-level:

$$Y_{ijk} = \gamma_{000} + \gamma_{010}(Tx_{jk}) + \gamma_{100}(Pretest_{ijk}) + e_{ijk}$$

Where γ_{000} is the mean post-test score and $\gamma_{010}(Tx_j)$ is the test of treatment group differences

from the comparison on the posttest, controlling for the student pre-test (i.e., γ_{100}). The student level variance (i.e., the variance of e_{ijk}) has constant variance σ^2 . If the treatment indicator is statistically significant, Hedge's g will be estimated. To control for multiple statistical tests being employed, we will be using Benjamini and Hochberg's (1995) Linear Step Up procedure to control for the False Discovery Rate.

We will explore moderation by student characteristics by including interaction terms between focal moderators (e.g., grade level, language status, AI literacy) and the treatment indicator. To address missing data, data will be analyzed to determine if they are missing at random (MAR) or completely at random (MCAR). For data that are MAR or MCAR, full information maximum likelihood analysis will be used. Non-ignorable missing data will be treated via methods identified by Demirtas (2005).

Integration of Qualitative and Quantitative Data. Our quantitative data will be used to point us to particularly effective, and ineffective, usage of PapyrusAI as shown by student outcomes. We will re-analyze trends seen in the quantitative data by considering the qualitative data for the applicable instructors and students to better understand why such outcomes may have occurred. Conversely, our qualitative data will suggest which high (and low) implementations observed are worth quantitatively analyzing separately as well as qualitatively in Year 4 through our case studies. Finally, both qualitative and quantitative data will inform iterations of the intervention, giving us the ability to look at a granular level to understand the types of interactions (by prompt, or type of assignment, or frequency) that seemed to lead to the highest outcomes and should be further supported by the intervention.

Power Analysis. We used the PowerUp tool (Dong et al., 2015) to calculate the minimum detectable effect size for our study (MDES calculator for a 3-level fixed effects blocked cluster random assignment design). Our analysis parameters were set as follows: Alpha .05, two-tailed, power .80, R_1^2 .40 and R_2^2 .05. Estimating 6 schools, 10 instructors in each, randomized across PapyrusAI treatment and BAU control, and 20 students each, our minimum detectable effect size is 0.258, which is within our expected size of treatment effect based on Graham et al. (2023). Research shows that teaching secondary school students to engage in increased planning, the provision of formative feedback, and the use of digital technology improves their written composition with effect sizes from .31 for digital writing tools to .49 for planning (Graham et al., 2023). Preliminary results by Co-I Hebert and AB member Graham in their IES funded meta analysis study of postsecondary writing suggests this effect size range will be similar for our population. We also expect a meaningful change in students' AI literacy greater than that found in the validation study, which consisted of a single task using a chatbot compared to our semester-long intervention, $\beta = 0.220$, $t = 2.093$, $p = .040$ (Jin et al., 2024).

Pilot of Promise Study: Cost Analysis

Aim 5: To estimate the cost of implementing the intervention. What is the cost of implementing PapyrusAI?

We will document the type and quantity of resources needed to implement the intervention, including the cost of AI (API costs, Amazon Web Services hosting costs) and technology access (percentage of use), amount of technology support utilized, and hours of PD over the course of the RCT using the Ingredients Method (Belfield et al., 2018). A major cost of the intervention will be the fees for AI access. These will be calculated by dividing the periodic fees by the number of students served. As for remaining costs, our Year 3 teacher surveys, interviews, and

classroom observations will gather information on the amount of device usage to allow us to calculate the percentage of use in treatment and control classes, and other program ingredients including the cost of PD and the possibility of induced costs.

We will consult with our partner colleges to gather cost data from their technology staff for local pricing data and personnel costs and use the CBCSE Estimator tool (Hollands et al., 2022) for national pricing information. Ultimately, we will triangulate our data sources and to the extent we have differences in the information we will describe a range of pricing or quantity. For any costs that are estimated, we will run sensitivity analysis to determine the impact of different assumptions. We will do this analysis at a college level, because both the technology support and purchasing are generally done at this level, and because the course curriculum is often determined at this level. Our analysis will compare the costs for the intervention to those of the control classes in the BAU condition. We will estimate technology costs (devices, support, internet access) on a percentage of use basis, and amortize the cost of devices over the estimated lifespan. Finally, we will provide a reference case analysis adopting a societal perspective, using national average prices and a 3% discount rate to calculate present values. Our work will follow the steps in the Cost Analysis: A Starter Kit (Institute of Education Sciences, 2020). We will report costs on a per-class and per-student basis.

Time and Training Needed to Adopt and Sustain Implementation. We will track the time and training of participating instructors in Year 3 (RCT). Training will be tracked through attendance logs of the PD and monthly meetings.

Aim 6: To develop an Implementation Support Guide. What do researchers and practitioners need to implement the tool in community college writing instruction?

Implementation Support Guide

The Center will document and analyze the requirements and recommendations for successful adoption and monitoring of the AI tool beginning in Year 2 as part of our analysis of usability and feasibility. We will draft the Implementation Support Guide (version 1) in spring of Year 2 so that it can be an instructional resource for our RCT treatment teachers in Year 3. This initial draft will also integrate findings from our exploratory study of generative AI tools, including the findings from interviews that help understand the necessary expertise, conditions, and resources for successful implementation of similar tools. We will include draft assessments to allow users to check both (a) whether they have the necessary expertise, conditions, and resources to implement the tool, such as necessary information and assessments to monitor implementation, enact a continuous improvement process, conduct preservice and in-service professional development, set up appropriate communication and data systems (e.g., Scaling checklist, Metz et al., 2020; see Appendix B-7), have the necessary implementation infrastructure (Appendix B-8), and ensure that financial and human resources are in place (Appendix B-9); and (b) how well they are implementing the tool based on performance goals, for example.

This first iteration of the guide will be a resource for the treatment teachers in the Year 3 RCT, during which we will determine fidelity to core components in treatment and get a clear understanding of the BAU control condition. As part of the RCT, we will solicit feedback from our treatment teachers. A revised guide (version 2) will be provided to the control teachers who choose to implement the tool in Year 4. At the end of the semester, we will ask instructors to provide feedback, and Co-PIs in our community colleges will also provide feedback on the draft guide and its usability. Revisions to implement suggestions will be made during Year 5 as we finalize the Implementation Support Guide for publication.

If the pilot study finds that the tool holds promise of beneficial impacts on postsecondary students' academic outcomes, the Center will further develop this work into a **practitioner-friendly implementation support guide** (version 3) that will help postsecondary institutions adopt and evaluate the generative AI tool. We will use our Year 4 case study teachers as well as our AB members to provide suggestions and help refine the guide for a wider audience of researchers and postsecondary institutions. Our case study teachers will also provide feedback on our assessments that are included in the guide and ensure they are both comprehensive and understandable to the relevant populations.

Our data and measures are described above in the development study as are the professional development and instructional resources that we plan to provide to treatment teachers. We will identify potential moderators of implementation through our qualitative work (e.g., instructor interviews, class observations) as well as the presurveys of instructors. We will assess, for example, if higher levels of AI literacy at the beginning of the semester are related to increased student outcomes using the regressions and models discussed above. We will review classroom artifacts, such as course syllabi and assignments, to determine if certain amounts or types of AI incorporation are correlated with higher student outcomes (e.g., coding for revision or planning use of the tool). Finally, our case study of 12 successful instructors in Year 4 will provide insight that will help guide this work and help us identify adaptations made in response to local context and organizational factors at the course and institutional levels. This work will provide insights into the **resources** needed and **barriers** to the adoption of such tools.

If the pilot study does not find evidence for the promise of beneficial impacts, the Center will release a more general guide based on its findings of the implementation needs and supports for the tool to inform those developing and implementing similar tools.

Ensure Safe, Valid, and Unbiased Results

We will prioritize the creation of safe, valid, and unbiased results for the intended learners. PapyrusAI will use OpenAI's Moderation API to ensure that AI output does not contain inappropriate language or content (e.g., hate speech, harassment, self-harm). In addition, API terms ensure that no student or instructor data will be maintained for model training. The instructional resources will provide students with clear usage guidelines, tutorials, and built-in supports that promote responsible and safe use. We will comply with all institutional review board requirements (see Human Subjects Plan), including participant informed consent.

We will also employ data analysis techniques to monitor and assess whether the accuracy and relevance of the AI-generated content differs meaningfully across student groups by race, language background, gender, ethnicity, or age. Key to this process is the use of NLP tools and machine learning algorithms that allow us to identify biases and disparities in response quality or feedback effectiveness (see, e.g., Nixon et al., 2024). If meaningful biases or disparities are found, we will adapt our system instructions and embedded prompts to attempt to address them.

Additionally, we will gather during our observations, focus groups, and interviews any perceived inconsistencies or biases in the AI's interactions. This direct input from users will supplement our data-driven approach, providing actionable insights that guide iterative improvements to the AI model. By embracing a cycle of continuous monitoring, feedback collection, and adjustment, we are committed to upholding a high standard of fairness and quality in the educational experiences facilitated by PapyrusAI, ensuring it serves as a reliable and unbiased resource for all students.

More generally, our Ethics Advisors will work to identify and address a broad range of ethical issues, including how instructors and students are informed and have opportunities to opt-

out. Following recommendations of NIST researchers (Greene et al., 2024) we will use the Belmont Report categories—respect for persons, beneficence, justice—to organize the sprawling list of ethical concerns raised in the ethics of AI in education literature (Adams et al., 2023). Evidence of potential ethical issues will be captured and reported to all PIs. For each issue, the Ethics Advisors will be involved in the development and review of the plan for mitigation or management. They will participate in the decision meetings that determine the enhancements to be evaluated in studies, offering recommendations on whether the resolution of ethical issues has been sufficient to allow the next stage of research to proceed.

Reducing Achievement Gaps

The proposed project is based on the premise that AI writing tools are flawed and fallible, due to their algorithmic biases and technical shortcomings (Birhane & Raji, 2022), but that they also have powerful affordances, especially for disadvantaged writers who disproportionately benefit from personalized tutoring. In addition, AI literacy is especially necessary for students who face a lack of access and participation in technology-related fields and are more likely to fall prey to misinformation and unfair decisions made by AI (Rospigliosi, 2021). Our goal then is not to produce an AI that performs perfectly, as that is an unachievable goal, but rather one that can meaningfully improve learning outcomes for all students, especially those in need.

In alignment with SEER standards, we will pay close attention to the needs of low-performing students at every step of the Center's research. During the research, we will note any differences that emerge and ensure that classes with at-risk students are included in the case studies. We will assess any differences in engagement and outcomes using descriptive measures in the early years and moderator analysis during the RCT. Throughout, we will seek to identify potential levers for tool improvement (such as in its interface, reading level, communication style, etc.) to address disparate outcomes.

Data Sharing and Open Science

Through his work as both a PI and founding editor of two prominent open access educational research journals, PI Warschauer has long been a leader in promoting open science and data sharing and Co-PI Tate has published several public datasets for the Lab. Research data will be anonymized and shared (see Data Sharing & Management Plan for details). All hypothesis-testing studies conducted will be pre-registered.



Training and Fellows Program

The WRITE AI Center will establish a comprehensive fellows program to build national research capacity for rigorous evaluation of generative AI implementation in postsecondary education. With an emphasis on writing instruction in postsecondary courses, the program will train 40 researchers, educators, and institutional leaders with the methodological skills necessary to evaluate AI tools' effectiveness, impact, and ethical implications. Drawing from MDRC's TANF Data Collaborative model (see Appendix B-10), this program will ensure fellows gain the skills

to assess AI tool implementation, measure learning outcomes, and conduct ethical AI evaluations in real-world educational settings.

To recruit fellows, direct outreach through conferences and social media will be complemented by targeted recruitment at minority-serving institutions, rural colleges, and under-resourced institutions. The competitive selection process will build cohorts that bring together postsecondary instructors, educational researchers, institutional researchers, EdTech professionals, and graduate students. Fellows will convene annually at an in-person stakeholder summit, where they will present research findings, participate in hands-on workshops, and participate in a hackathon to implement best practices for AI evaluation.

Each fellow will receive a travel stipend for one trip and participate in structured activities including one introductory session, six modules (webinars modeled after those delivered in MDRC's data collaboratives) and regular office hours with research faculty. The fellows program will operate through a structured, virtual-first approach over three years to ensure accessibility for participants from diverse institutions. Year 1 will focus on planning and needs assessment, followed by Years 2 and 3 implementing the fellowship program with two cohorts of 20 fellows each. At its core are intensive virtual minicourses, delivered live and recorded for asynchronous access through MDRC's technical assistance platform.

Core Training Modules

The fellows program comprises six integrated training modules (see Appendix B-11), each designed to build specific capacities for rigorous AI evaluation research while connecting directly to the Center's ongoing research activities. All instructional sessions will be led by MDRC and affiliated experts.

- ***Evaluation Frameworks for AI in Postsecondary Instruction***, including methodological approaches for measuring learning gains, implementation fidelity, and student engagement in AI-supported instruction.
- ***Causal Research Design for AI Evaluation***, including through randomized control trials, A/B tests, factorial experiments, and multi-armed bandit approaches specifically adapted for AI evaluation, and emphasizing SEER-aligned practices.
- ***Measuring Engagement and Tool Behavior***, including advanced techniques (sequence analysis, Sankey flows, clickstream data) for assessing tool behavior, evaluating AI-generated feedback, and linking user engagement metrics with learning outcomes.
- ***Research Data Collection and Management***, including practical skills for managing large-scale AI-generated data, integrating multiple data sources, ensuring data quality, maintaining compliance with privacy and FERPA regulations.
- ***Mixed-Methods Research for AI in Postsecondary Education*** to develop comprehensive understanding of AI implementation in postsecondary settings.
- ***Ethical Considerations and Bias Mitigation***, including how to identify potential biases in AI-generated feedback, develop mitigation strategies, and create frameworks for ensuring ethical implementation.

As the program's capstone, fellows will participate in a dynamic three-day AI in Education Design Sprint hackathon that reimagines how artificial intelligence can enhance writing instruction in higher education and emphasizes conceptual innovation over technical implementation, thus enabling meaningful contributions from participants regardless of their engineering background.

Each fellow will gain access to an extensive online resource library containing evaluation frameworks, code repositories, and sample protocols developed through the Center's research

program; ongoing technical assistance from MDRC's AI and educational evaluation researchers; and a collaborative workspace to co-develop research projects and share findings.

To ensure sustainability, an active alumni network will support ongoing collaboration and mentorship among program participants. The Center will develop and maintain a publicly accessible repository of training materials and evaluation frameworks, ensuring these resources remain available to the broader research community. Long-term partnerships with postsecondary institutions will be formalized through memoranda of understanding, establishing sustainable structures for ongoing AI evaluation research. The Center will also develop policy recommendations to guide responsible AI tool implementation in writing instruction, helping institutions build internal capacity for evidence-based decision-making about AI adoption.

National Leadership, Capacity Building, and Outreach Activities

The WRITE AI Center will carry out national leadership, capacity building, and outreach activities to promote the Center's national visibility, engage with the fields of research and practice, and allow it to function as a trusted source of scientific research in the use of generative AI to improve teaching and learning. Our work will focus on communicating actionable information and research findings, bringing stakeholders together, and developing resources to support the use of generative AI in postsecondary instruction.

The primary focus of our national leadership, capacity building, and outreach activities will be on the teaching and learning of writing, and we will also place special focus on those involved in community college contexts. However, we also recognize the advantages of reaching across these boundaries—for example, key community college leaders we intend to reach are often focused on instruction more broadly rather than in a single domain, or may benefit from learning together with people in other types of postsecondary institutions—so our leadership activities will in part address broader issues of postsecondary learning with generative AI.

The following is a summary of the major initiatives that we will carry out in our three main areas of work. In a number of cases one of our Center's partner organizations will take the lead (e.g., the Digital Center for webinars and professional development and MDRC for podcasts) drawing on their pre-existing infrastructure for the activity, thus best leveraging grant funding; however, in all cases, outreach will be carried out by all partners nationally through Center and partner websites, email lists, and social media to reach the broadest possible audiences. Past experience in national leadership suggests that a mix of planned and opportunistic activities is necessary. Planned opportunities ensure that necessary information is disseminated and engagement occurs. Opportunistic activities respond to requests to our Center from our audience.

1. Communicate Actionable Information and Research Findings

A project **website** will be developed that focuses on the use of generative AI for postsecondary teaching and learning and serves as a repository and amplifier of all of the leadership work of the Center. The website will make the project's goals and activities transparent to multiple audiences, researchers, policymakers, educational leaders, educational practitioners, parents, students, media, and the general public and provide a central hub for our communications. The website will be modeled on the Digital Learning Lab, Online Learning Research Center, Pens & Pixels, and WRITE Center websites developed by Tate and Warschauer to disseminate findings from previous grants, seen by thousands of visitors a week. It will include pre-prints, summaries of best practices for instructors and students, annotated bibliographies, downloadable versions of our videos, blog postings about our research findings, information on our webinars, downloadable information from our training materials and curricula, plain-English Executive

Summaries of research findings, both from the Center and others, describing what is working, for whom, and in what ways to use generative AI to improve writing instruction.

We will organize monthly educational **webinars** targeted at educational leaders, educators, and tool developers. Webinar hosting and organization will be carried out by the CCC's Online Network of Educators (see Appendix B-12 for the numerous spring 2025 webinars hosted by the group on AI in postsecondary instruction). The webinar series will feature presentations by the CCC Digital Center's 10 annual faculty fellows, chosen from across the CCC system for expertise on generative AI-based instruction, and will include topics such as the relationship of generative AI to academic integrity, students' AI literacy development, and feedback on student writing. Webinars will periodically focus on institutional leadership for adopting and deploying generative AI for teaching and learning, drawing on the expertise of ACE in selecting speakers and reaching the appropriate audience. All webinars will be recorded for broad access.

We will organize quarterly **podcasts** on empirical research on generative AI in writing as part of MDRC's monthly *Evidence First* podcast series. Each podcast will feature researchers who have carried out empirical studies and will include topics such as the ethics and biases of AI writing detection tools, the impact of generative AI use at scale on students' discussion posts, the relationship of ChatGPT use to student writing performance, and comparisons of the quality of generative AI compared to human scoring and feedback.

We will develop annual **briefs**, published on our website, on important topics related to teaching and learning of writing with AI. Our first will be on issues of ethics, bias, privacy, and academic integrity in relationship to writing and AI; subsequent briefs will depend on needs that emerge from our discussions with stakeholders.

We will **publish findings** from the Center in a wide range of journals and academic conferences directed at both researchers and practitioners in the fields of composition, education, higher education, educational technology, and AI and learning. We will also use social media as appropriate (e.g., BlueSky, LinkedIn) depending on the audience. Cognizant of the importance of getting out research findings rapidly in this rapidly changing field, all of our papers will first be published as preprints. To align with principles of open science and replication, all hypothesis-testing studies will be preregistered and all data from our research will be made publicly available (see Appendix A). In addition, we will publish a short monograph overview of research on generative AI for writing in the Cambridge Elements series on Generative AI in Education, edited by PIs Warschauer and Tate.

2. Bring Stakeholders Together

We will participate annually in the **Futures Summit**, an annual meeting hosted by the Digital Center in Silicon Valley, bringing together policymakers, practitioners, researchers, and tool developers on the topic of AI in higher education, with a focus on community colleges. A total of 400 participants joined the inaugural Futures Summit, held in Cupertino, California in September 2024; the Digital Center intends to increase that to 800 participants at the next Futures Summit to be held at the Santa Clara Convention Center in September 2025. The WRITE AI Center will hold a series of special sessions in both a pre-summit gathering and during the main event to share and discuss the latest research findings on generative AI for the teaching and learning of postsecondary writing, identify priority avenues for future research, help AI tool developers understand research-based approaches to improving student writing, and discuss ways of influencing educational policy and practice with evidence-based guidance.

We will bring these conversations to other venues that serve diverse higher education audiences through workshops and panels at the State Higher Education Executive Officers

Associations, EDUCAUSE, the WICHE Cooperative for Educational Technologies, and Achieving the Dream.

We will also organize two stakeholder convenings, one in Year 2 and another in Year 4. Leveraging ACE's network and reach within the U.S. postsecondary sector, we will bring together 75 to 100 institutional leaders, subject matter experts, thought leaders, and policy makers across the country for a one-day, in-person meeting on AI for postsecondary writing. Program contents will be informed by the exploratory study, as well as other components of this project, and will aim to inform institutional policies/practices, as well as public policy that increases the research-based, ethical, and responsible use of AI to improve student writing, learning, and digital literacy outcomes for college and career success.

3. Develop Resources

In addition to a publicly available and searchable database with a web-friendly user interface (see discussion in exploratory study above), we will develop and share additional resources to build the capacity of postsecondary institutions to use scientific research on generative AI for improving instruction and learner education outcomes.

Building on earlier work by project PIs in developing evaluation frameworks for digital learning (e.g., the Online Course Quality rubric, Xu, Li, & Zhou, 2020), we will develop an Evaluation Framework for AI in Postsecondary Instruction by spring of Year 4..

We will develop (in Year 1) and deliver (in Years 2-5) a professional development certificate program for postsecondary educators on the teaching of writing with genAI offered to educators across the country through @ONE, similar to Certificate in Online Teaching & Design it offers, and covering topics such as the nature of AI-assisted writing, selecting among AI tools, design courses incorporating AI, appropriate guidelines for student use of AI, ensuring ethical use of AI, and new forms of writing assessment in the AI era.

Management and Institutional Resources

Management Structure and Procedures

PI Warschauer will provide oversight and leadership to the overall Center. MDRC will take the lead in the exploratory study, assisted by ACE in survey implementation and UCI in case studies of writing platform use. UCI will lead in the implementation and pilot study and development of the implementation support guide with MDRC taking responsibility for the cost analysis and for serving as an external monitor on the design and evaluation of the pilot study. As for the national leadership activities, California Community Colleges via Foothill De Anza will lead the Futures Summit, the webinars, and the professional development certificate program; ACE will lead innovation labs and assist with national dissemination; MDRC will lead the podcasts and evaluation framework; and UCI will lead the website, briefs, and searchable database. MDRC will lead the research training activities. See Appendix B-14 for an overview of the Center's management structure.

Each phase of the project will be implemented following the Plan-Do-Study-Act framework (PDSA; Langley et al., 2009; Wilcox & Zuckerman, 2019; see Appendix B-13). We will collaboratively plan (set goals, compare practices, select use cases), do (make an action plan, integrate PapyrusAI and AI literacy into ELA classes), study (monitor and evaluate the progress, analyze findings), and act (implement the intervention). The PDSA will be conducted in a cyclical manner to iteratively improve the products under development.

Regular ongoing meetings will be held of the PIs, for ongoing oversight, as well as of the various project teams, to coordinate activities and share emerging requirements, insights, and

feedback. PIs will meet with the advisors quarterly throughout the project to ensure regular monitoring and input, divided between at least one full group meeting annually and subgroup meetings throughout the year.

All project documents will be stored on a shared Google drive, with folders and subfolders used for ease of sharing materials. A system of running agendas and notes will be used via Google docs for the overall PI meetings as well as for the project team meetings to ensure comprehensive and accessible note taking. Zoom will be used for video communication. Email and a Slack channel will be used for ongoing communication. The postdoctoral researcher will devote one quarter time to overall project coordination, and will use Asana as a detailed project management tool to keep track of tasks and activities, responsibilities, status, and milestones.

Resources

University of California, Irvine. The UCI School of Education is one of the most highly rated education research schools in the world with particularly strong faculty groups in digital learning and language and literacy development. The School is launching a new master's program in AI and Learning Analytics which will be one of the first such programs in the United States. The School has a dedicated Business Services Office and IT team, supported by the Office of Research Administration and Office for Information Technology on campus. The School has 40,321 assignable square footage that includes seminar and meeting rooms for research activities. Dedicated project lab space is available within the Digital Learning Lab, a 2000 square foot lab across the street from the main School of Education building. The Digital Learning Lab also includes office space suitable for all postdoctoral researchers and GSRs who will be hired, as well as undergraduates working on the project. Finally, it includes two video conference rooms, one seating 5-10 people and one seating 30 people, that can be used for all research and advisory board meetings. The School has successfully led many large federally-funded projects, including the IES-funded WRITE Center for Secondary Student Instruction.

MDRC. MDRC is well-positioned to oversee the exploratory study and the research training and evaluation work of this center, based on a 50-year history of conducting large-scale rigorous, reliable demonstrations and evaluations of interventions intended to improve the lives of low-income people. MDRC's Postsecondary Education (PSE) group has conducted numerous experimental and quasi-experimental evaluations of the implementation and impacts of strategies and interventions intended to improve the academic success of low-income and academically underperforming students. MDRC's PSE group also conducted a complex national survey of public and private postsecondary institutions, collecting information from over 850 institutions about their student assessment and instructional practices for developmental education. The team provides evidence-based technical assistance to help colleges and higher education agencies implement proven programs and practices. The PSE group's mission is to generate evidence and provide technical assistance focused on postsecondary students and students transitioning to postsecondary education, to promote positive student outcomes. The CDI group provides data analytics, technical assistance, and support services. It has developed and applied methods for rapid feedback for program improvement, including rapid-cycle tests, A/B tests, and machine learning, and conducted summative evaluations following formative feedback studies. It has also studied AI based platforms using methods similar to this proposal with a focus on engagement, disparities, A/B 33 optimization testing and efficacy. CDI uses a data-informed, human-centered approach to improving programs and user experience.

ACE. For over a century ACE has driven transformative change in American higher education, with achievements that include launching the Education Testing Service and the GED; helping author and push for passage of the GI Bill; leading efforts on the Higher Education Act of 1965, including its key provisions to expand higher education access to low-income and minority students. For more than 60 years, ACE has supported adult learners and the higher education institutions that serve them. Building onto these legacy efforts, some of the active initiatives at ACE continues to strengthen its capacity, reach, and influence as the Council that leads the postsecondary sectors to serve more learners & facilitate their successes. A recent area of emphasis for ACE has been to address the complex challenges and opportunities that emerging AI technology presents for the sector. Coming out of the June 2024 convening at the White House, AI in Postsecondary Education Working Session: Advancing AI Together, organized by the U.S. Department of Education, it is evident that higher education institutions must organize themselves more proactively and swiftly to foster sustainable collaboration. Building onto this convening, ACE has launched the Global Data Consortium (GDC) as a critical coordinating body for AI innovations in postsecondary teaching and learning that will: 1) connect and mobilize discrete, niche networks of experts as field-wide efforts; 2) serve as a steward of learner information across the sector; and 3) facilitate continuous capacity building for individual institutions and the sector. This spring, with funding support from the Gates Foundation, ACE will be starting the exploratory & planning exercise for the data sharing infrastructure.

Foothill-De Anza Community College District. Foothill-De Anza in Silicon Valley, California, has a track record of helping lead a number of statewide and local CCC grants, most notably the California Virtual Campus-Online Education Initiative (CVC-OEI), the Online Career Technical Education (CTE) Pathways grant programs (comprised of 70 college subawards), the CVC Course Exchange, and the Online Network of Educators (@ONE), which has more than a decade of experience developing professional development training courses and webinars for educators across the state and nation. Foothill-De Anza was recently selected as operational lead for the new Digital Center, a strategic initiation to supporting Vision 2030 for the California Community Colleges, including leading in innovation, harnessing the power of generative AI, public and private technology partnerships, and transforming the system's ability to serve students as they move into an AI-powered workforce. The Digital Center is designed to enable the system to quickly adapt and innovate, with a fundamental component involving prototyping, piloting, and assessing technological innovations to scale effective partnerships, strategies, technologies, and practices across California community colleges. The Digital Center serves as a hub for evaluating new technologies and partners based on their impact on student learning, cost-effectiveness, and alignment with CCC values.

Virginia Community College System. The VCCS is a network of 23 community colleges serving students across the Commonwealth of Virginia. Established in 1966, VCCS provides affordable, accessible, and high-quality education, including associate degrees, certificates, workforce training, and transfer programs to four-year universities. With over 250,000 students enrolled annually, the system plays a crucial role in workforce development, economic mobility, and lifelong learning. VCCS institutions collaborate with local industries and government agencies to align programs with Virginia's economic needs, particularly in fields like healthcare, cybersecurity, and advanced manufacturing.

Personnel

PI Mark Warschauer (25%), Distinguished Professor of Education at UCI with appointments in Informatics, Psychological Science, and Language Science, and Director of the Digital Learning Lab, one of the most widely cited researchers in the world on digital and AI-based writing, and has served as PI on \$30 million of funded research, including as co-PI for the IES-funded WRITE Center (2019-2025). Warschauer will be responsible for overall management of the Center and for coordination with partners and advisors. **Co-PI Di Xu (8.3%)**, Professor of Higher Education Policy at UCI with expertise in improving and evaluating learning outcomes in community colleges through innovations in digital learning, will lead the evaluation of the pilot study). **Co-PI Tamara Tate (16.7%)**, Project Scientist at UCI, Associate Director of the Digital Learning Lab, and a PI of four previous grants on AI-based writing, will lead the Development study and Implementation Support Guide and help coordinate the Leadership activities. **Co-PI Alexander Mayer (10.6%)**, Director of Postsecondary Education at MDRC, will lead the exploratory study. **Co-PI Richard Hendra (10.8%)**, Director of the MDRC Center for Data Insights, will lead the Research Training activities. **Co-PI Jory Hadsell, Ed.D.**, Vice Chancellor of Technology and Chief Technology Officer of the Foothill-De Anza Community College District and Executive Sponsor of the Digital Center, will help lead the Leadership and Outreach activities and oversee implementation of research activities in the California Community Colleges. **Co-PI Justin Schultz (5%)**, Director of Strategic Planning and Operations for the Foothill-De Anza Community College District and Fiscal Administration, Project Management and Operations Lead for the Digital Center, will help coordinate the Leadership and Outreach activities and oversee implementation of research activities in the California Community Colleges. **Co-PI Hironao Okahana, Ph.D. (11.1%)**, Vice President & Executive Director, Education Futures Lab, American Council on Education, will serve as the project director for ACE, and provide overall leadership to all aspects of ACE's participation in the Center. **Co-PI Sheri Prupis (7%)**, VCCS Director of Teaching & Learning Technologies, will oversee implementation of research activities on the VCCS campuses.

Other key personnel at UCI include **Michael Hebert (4.2%)**, UCI Associate Professor of Education, Director of the Writing Project, and PI of a meta-analysis on writing interventions in postsecondary education, who will lead the work on measures used in writing evaluation; **Cristina Lopes (2.1%)**, Chancellor's Professor in Informatics and director of the Master of Software Engineering Program, who has expertise in development of generative AI-based learning tools and will advise on development of PapyrusAI; **Beth Harnick-Shapiro (2.1%)**, Lecturer in Engineering, who has expertise in AI for disciplinary writing and will provide leadership in that area; and **Penelope Collins (4.2%)**, Associate Professor of Education who has expertise in developmental writing, including for students with disabilities and language challenges, will advise in that area.

Our **Writing Advisors** are **Charles MacArthur**, Professor Emeritus at University of Delaware and an internationally renowned expert on writing in the community colleges; **Steve Graham**, Regents and the Warner Professor at Arizona State University (ASU) and one of the world's top researchers on best practices in writing instruction; and **Anna Mills**, community college writing instructor who is lead advisor for the AI writing platform MyEssayFeedback and a member of the MLA/CCCC Joint Task Force on Writing and AI. Our **AI-based Learning Advisors** are **Ryan Baker**, Director of the Penn Center for Learning Analytics and co-PI in charge of research capacity building in the IES funded Using Generative AI for Reading R&D Center, and **Danielle McNamara**, Executive Director of the Learning Engineering Institute at ASU and developer of Active Learning at Scale, an IES-funded AI-based learning platform for

postsecondary students. Our **Ethics Advisors** are **Renzhe Yu**, Assistant Professor at Teachers College of Columbia University and director of the Advancing Equitable Technology & Algorithms in Society lab, and **Fred Oswald**, Professor and Herbert S. Autry Chair in Social Science and National AI Advisory Committee member with expertise on the ethical and responsible use of generative AI in higher education, including approaches to protecting privacy and combating cognitive biases.

Other Personnel

Houston Community College Vice Chancellor of Instructional Services Miguel Ramos will oversee implementation of research activities at HCC. ***Director of Communications for the Digital Learning Lab*** Sabrina Look will contribute to the national communications of the Center. ***Software Engineer*** Kristi Werry, who has led development of Papyrus AI since its inception, will implement changes to PapyrusAI and also develop and maintain the searchable archive of generative AI writing tools. ***One Postdoc and three Graduate Student Researchers*** will be hired to help carry out all aspects of the research, including development of the AI writing tool, prompts, and instructional resources; data collection, management, and analysis; and publications and presentations. ***Undergraduate Researchers*** will be involved from the Career Pathways for Researching Learning and Education, Analytics and Data Science program, an IES-funded training project to develop interdisciplinary expertise in education data science.

Objectivity and Transparency

The principal investigators on the project are long-term leaders in the open science movement. PI Warschauer has founded and led two open access journals, *Language Learning & Technology* (1996-2003, 2011-2014) and *AERA Open* (2014-2022), which implemented open science principles, such as inclusion of registered reports and data sharing requirements. The Digital Learning Lab led by Warschauer and Tate publish all their grant proposals, instructions, and research data, and preregister all hypothesis-testing studies. All of these open science principles will be continued in the center. PapyrusAI is a noncommercial tool developed under a grant to the University of California from the National Science Foundation and made publicly available through an open source MIT License. To ensure that the evaluation of it is done in an objective matter, in addition to the above mentioned steps of preregistration and data sharing, all research procedures and analyses will be approved by MDRC, which is not connected to the development of the tool, and random assignment of participants and the costs analysis will also be conducted by MDRC. The above mentioned Center advisors will (1) assess whether the project is making satisfactory progress towards its goals, (2) recommend reasonable and evidence-based adjustments to project plans, (3) oversee that the key personnel maintain their objectivity in their research, (4) assess the overall quality of the research that is produced as a result of this project, and (5) attest to the integrity of any outcomes that stem from the project, with the entire board meeting annually and each subgroup providing advice and oversight within their area of expertise (writing instruction, AI-based learning, and ethical use of AI in postsecondary instruction) throughout the year.